

Workplace Cyborgification in Maritime Onshoring: An Ethnographic Case Study

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Abstract

This article examines an emerging paradigm of “maritime onshoring” in the offshore petroleum industry, focusing on work practices enabled by remotely operating subsea ROVs from onshore locations using low bandwidth 4G technology. Traditionally, ROVs are piloted offshore during inspection, maintenance, and repair operations. Based on a cognitive ethnography of an onshore Remote Operating Center, we analyze how pilots and inspection engineers assess deep-sea infrastructure. While initially motivated by questions about how maritime onshoring reshapes technology-intensive maritime workplaces, this study argues that this process instantiates a key phenomenon we call “workplace cyborgification” — where ROC members encounter, harness, and assimilate digitally mediated action-perception loops to accomplish work. Through an extended sensorium and tightly coordinated co-operative actions, pilots, inspectors and ROVs operate as human-technology hybrids, seamlessly conducting delicate work at great distances in the subsea environment. From the interactional analysis, we draw out three conceptual and empirical implications for the CSCW literature.

CCS Concepts

• **Social and professional topics** → Professional topics; Computing and business; Computer supported cooperative work; • **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; Collaborative and social computing; Collaborative and social computing design and evaluation methods; Ethnographic studies.

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1 Introduction

Control rooms have long been considered ethnographic windows for examining the nature of computer-supported cooperative work. In Suchman’s classic formulation, control rooms present technology-intensive sites for overseeing work at a distance, across space and time, as “centers of coordination” [1]. Having seen a paucity of studies on this area in the last decade, recent CSCW-scholarship have made the case for “moving back into the control room” [2]. This study answers that call by studying a novel Remote Operating Center (ROC) enabling “maritime onshoring” — the relocation of ROV operations from sea to land in Norwegian maritime oil and gas industry. This transition is facilitated by new, sophisticated network technologies for remotely controlling ROVs for subsea work. The term ‘cyborg’ originates from cybernetics — the study of control and communication in both organisms and machines — highlighting the intricate feedback loops shaping human-technology interaction. In this study we move back into the control room by reconceptualizing it through the lens of “workplace cyborgification”. Acknowledging the diminishing divide between the virtual and physical in maritime oil and gas work [3], we draw on Andy Clark’s seminal idea about humans as historically being “natural-born cyborgs” [4]. Here, we emphatically use the term cyborgification, not in a metaphoric sense, but in Clark’s analytical spirit to describe some of the concrete processes by which workers integrate digitally mediated action-perception loops into their daily practices. These are fast and tightly coupled, creating immediate feedback between operator action and system response. In the technologically dense environments like the control room studied here, this is a process that reconfigures labor, skill, and agency. Workplace cyborgification thus refers to a situated, evolving integration of workers and digital systems — one that enables new forms of action. Here, technology becomes part of the workers’ problem-solving environment, enabling remote enactment of subsea work. Yet this is not a permanent state, but a dynamic one: when workers disengage from the technological interface, such as the ROV instrumentation, the cyborg condition may take on other forms or cease altogether. We argue that the interesting question is not whether ROC pilots and inspection engineers *are* becoming cyborgs, but *when* they take on cyborg properties, and under *which* conditions. This article should thus be read as an example of what Chalmers’ calls “*de novo* conceptual engineering” [5], whereby a new concept (‘workplace cyborgification’) is coined and implemented in the context of an ethnographic study¹.

¹Chalmers views conceptual engineering as comprising three main efforts: the design, evaluation, and implementation of concepts, each having sufficient scholarly value in its own right.

2 Background

Classic CSCW studies position control rooms as “centers of coordination” [1], managing work across space and time. While sharing characteristics with control rooms studied previously (e.g., managing remote, hazardous environments), the ROC we describe exhibits some key differences. First, it involves a distributed setup, coordinating onshore operators with offshore assets and personnel via digital networks. More fundamentally, its activities go beyond mere monitoring; operators and technologies become tightly integrated, forming hybrid cognitive systems with technology, enabling action-at-a-distance. Our concept of workplace cyborgification centers around this situated, dynamic integration. We align the study with cyborg anthropology [6, 7] in viewing humans as technologically entangled beings, where minds and bodies are continuously reshaped through tool use. As our ethnography shows: in the ROC, pilots and engineers develop new sensory-motor relationships with ROVs, digital interfaces, data streams, and manipulators, embodying hybrid entities for cognition and action distributed across vast distances. This process of distributed cognition and action reconfigures perception, agency, coordination, labor, identity, and skill, challenging traditional notions of control room work. It also foregrounds gaps between plans and situated action inherent in complex, technologically mediated work. Importantly, the kind of cyborgization described here is not about metaphorical or speculative futures, but about how labor gets reorganized via new extensions of human capabilities. As maritime onshoring reshapes the conditions of subsea industry, it exemplifies a lived, material transformation of work. These reconfigurations should be highly salient for CSCW researchers.

2.1 Situating the case in the broader landscape of control room studies

In some important ways, the ROC studied here aligns with studies of work in distributed environments, sometimes conducted over large (even interplanetary) distances [8, 9], where direct human presence is impossible for reasons of practicality, as well as safety concerns. For instance, our informants occasionally referred to the subsea environment as a place ‘not fit’ for humans. The work being conducted exists because of the mundane fact that things break down, particularly in the harsh subsea environment. This is reflected in the acronym for this type of work, usually referred to as “IMR”: inspection, maintenance and repair. This is also acronym used to classify the kind of vessels that host the ROVs. Inspection, maintenance and repair are words describing important types of work in modern society. The latter, for instance, provides a starting point for thinking about human relations with the material environment, both philosophically and ethnographically. Paraphrasing a recent collection of “repair work ethnographies” [10], terms like repair—alongside ‘inspection’ and ‘maintenance’—draws attention to a “common field of action”, including diagnostic work necessary, procedures related to finding solutions, and physical interventions, that help achieve desired results.

From the ROC, one can observe that repair takes place on three, interconnected levels. First, repair of underwater installations involves the maintenance and restoration of subsea infrastructure,

where ROV pilots and engineers navigate harsh conditions to assess damage and execute robotic interventions. Second, there is repair of communication breakdowns, a key feature of human sociality and an interface between interaction and cognition [11]. Disruptions between remote operators, offshore personnel, and automated systems, must be addressed, which require adjustments in interactional practices to restore intelligibility, between actors inside and outside the ROC. Third, repair of human-technology interaction involves troubleshooting failures in the operator-ROV interface, from fixing faulty probes to rebooting computer equipment, ensuring a smooth operation. Such layers of repair work, at quite different conceptual levels, highlight the entangled nature of material, social, and technological work in the ROC, where sustaining operations depends not only on fixing physical faults but also on maintaining communication and ensuring the proper human-machine integrations.

Additionally, the ROC in question differs from classic conceptions of control rooms in other notable ways. For instance, unlike control rooms that centralize operations in a single location within a bounded, physical environment, the ROC studied here is distributed and operates across multiple nodes, as remote operators manage and execute tasks onshore, by coordinating with sea-based personnel onboard ships, offshore rigs and other actors in the sub-sea environment, both human and nonhuman. This is enabled by digital connectivity, low bandwidth 4G networks, that makes it possible to perform tasks, enabling IMR work to be initiated and coordinated remotely from land. Additionally, the shift from sea to land changes the operator’s work environment. In theory, this enables a more flexible workforce mobilization that allows operators to support multiple projects simultaneously, without being co-located in the same ROC. Yet, the ROC does not merely present a site for remote monitoring, but a new social environment where relationships between people and technology differs from the one at sea, a dynamic space where operators and technology jointly are capable of action at great distances in the subsea world. By integrating remote sensors for vision and data streams for umbilical equipment that teams of pilots and inspection engineers can act through, including situational awareness tools, the ROC reflects the evolution of control rooms from bounded monitoring stations to distributed systems of action. The ethnographic materials we now turn to thus highlights ongoing processes of cyborgification in modern control rooms, where technological augmentation reshapes operators’ skills and roles in novel settings.

3 Notes on method

This study employs “cognitive ethnography” [12], an approach for examining how cognitive processes are distributed across people, tools, and environments in real-world activities. It is set within a broader research project on the automation shift in maritime oil and gas. Fieldwork took place between 2022 and 2024 at the ROC of a Norwegian firm specializing in remote maritime operations. Data collection involved site visits, participant observation in the ROC, partner workshops, field interviews, and video recordings of workplace activities. We adopted an ethnomethodological sensitivity to understand the practical methods operators used to make sense of and organize their work in this technology-dense environment



Figure 1: Overview of the main workspace in the ROC. ROV pilots to the left, inspection engineer on the right.

[13]. Multimodal video analysis allowed for fine-grained examination of verbal and non-verbal communication, tool use, embodied coordination, and the sequential unfolding of human-machine integration in real-time practice. This approach reveals the situated and ‘cyborgified’ nature of onshored, CSCW work in maritime oil and gas.

4 Findings: Cyborgification in the remote operating center

Let us now step into the remote operation center (Figure 1), commonly known as the ‘ROC’, and examine a slice of the activity in this “cultural-cognitive ecosystem” [14]. The dimly lit ROC hums with activity as two pilots and an inspection engineer manage an ROV hundreds of kilometers away and hundreds of meters below the surface. Operators sit before a mosaic of screens displaying camera feeds, telemetry data, task plans, and communication channels, faces illuminated by their glow. Two pilots sit on the left, while an inspection engineer sits on the right, overseeing and video-documenting that tasks follow the 31-page ‘riser inspection task plan’. The footage and audio is submitted to the chartering company’s oil and gas quality management system. Figure 2 illustrates a typical and mundane one minute segment of inspection work.

4.1 How speech acts and ostensive communication reveal cyborgification in practice

Let us focus on the pragmatics of the inspection work as illustrated in Figure 2, and what may seem like simple speech acts in the sequence, and how they illustrate cyborgification in practice. Here, we highlight the emergence of laminated, “co-operative”, context-dependent interactional accomplishments [15], as the team

integrates multiple semiotic resources, build on each other’s contributions, and use the structure of the interaction as a resource for further coordination. Due to limited space, we shall focus on two types of cyborgenic resources that are prominent in the one-minute sequence:

Extended sensorium for remote agency: Pilots utilize ‘chairs’, formally known as the ‘topside control system’ with multiple joysticks, touch screens, and buttons to control the ROV with minimal latency. Visual feeds from ROV cameras become extensions of sight, allowing navigation and focus on critical details. Telemetry data (pressure, depth, orientation) is integrated into a sensor-fed intuition of the remote environment. Powerful manipulators (like the Titan 4 “T4” arm), controlled via replica master joysticks, translate remote intent into precise mechanical action, extending dexterity across vast distances. Pilots seamlessly inhabit the ROV through transparent and seamless sensory immersion [4]. Telemetry data transforms sensor fed intuitions, ROV manipulators (guided by a joystick known as the replica master arm), extending pilots’ limbs to enable precise actions, limited only by the pilot’s eye, according to the needs of inspection engineers. The team becomes a human-technology symbiont with action and cognition distributed across the ROC, 4G network and ROVs.

Speech-action loops in ROV operation: In the first line, the instructor (right) initially asks the pilot flying in Norwegian, before switching to English for the voice recording and raising the microphone. Work unfolds through tight coordination between the inspection engineers, pilots and ROV, where particular “speech acts” [16] function as cybernetic commands, translating human intention into ROV action. The engineer directs the inspection based on detailed task plans while narrating observations for documentation. Communicative acts structure the workflow. Interactional modalities, include: 1. Directives and commissives: The engineer

1 Inspector: Yeah, should we go to the opposite side? [In Norwegian] Pilot: Yeah

2 Inspector: Going to the opposite side here, now

3 Inspector: Going up to the second pass

4 Inspector: Going to the southside

5 Inspector: Going to take a close look at this attachment here, on the south side.

6 Inspector: [Inaudible] should be placed on this tether, this location.

7 Inspector: Going to zoom out again

8 Inspector: Going to follow up to this vertical clamp



Figure 2: Transcript and visual exemplifying directive and embodied action illustrative of workplace cyborgification. Images represent the rightmost part of Figure 1. Pilots’ responses to the inspection engineer’s requests are visible from the camera feed on the monitor.

issues movement requests (e.g., Line 1: “should we go to the opposite side?”), and the pilot both verbally confirms (Line 2: “yeah”) and immediately executes the maneuver. These speech acts are more than simple instructions or acknowledgments; they extend the inspector’s sensory apparatus into the ROV via the pilot, with minimal delay. Although the inspector remains stationary, a bodily shift effectively occurs within the underwater environment, enabling further exploration. The directive (“should we go to the opposite side?”) and the commissive (“yeah”) blend seamlessly with the ROV’s movements, collapsing the boundary between verbal agreement and physical execution. The pilot’s verbal confirmation is embodied through real-time action, with latencies as low as tens of milliseconds, illustrating the immediacy of the action-perception loops central to cyborgification. 2. Representatives and ostension: The engineer also narrate ongoing actions “going to the opposite side here, now” (L3) and “going to zoom out” (L2) to both enact the pilot-ROV-assembly and document activities. This explicit narration ensures clarity of intention, tightening the alignment between the inspector, pilot, and ROV. 3. Language switching: Switching between Norwegian (informal, within-team coordination) and English (formal documentation for official recording) signals a shift into accountable, institutionalized labor bilingual practice embeds cyborgenic work within broader bureaucratic systems of documentation, surveillance and accountability, such as the quality system used to document subsea inspections.

5 Discussion: Workplace Cyborgification and the Implications for CSCW

The interlocutors are what Clark calls “electronic virgins” as they are sporting no silicon chips or implants [4]. Yet we argue that ROV-pilots and inspection engineers exhibit cyborgenic qualities in a substantive way. Our analysis demonstrates how maritime on-shoring through the ROC can be seen as workplace cyborgification, as the ROVs become cybernetic extensions of human agency. The ROC is not merely a place of oversight but a site where operators enact remote work through deep integration with technology. The ROV and its interfaces extends human sensing, cognition, and action, mediated through intricate communication and coordination practices as those described in the sequence above. These examples of cognition and action are deeply intertwined with the ROV’s systems, in ways that go beyond the cyborg as a metaphor. Participants integrate multiple semiotic resources A key aspect of workplace cyborgification is the emergence of the tight action-perception loops we have described above, where operators’ actions are immediately linked to changes in their perception of the remote environment, and vice versa. For example, the pilot’s joystick movements (action) directly and immediately alter the camera view and ROV position (perception). ROV technology extends sensing supported by video and telemetry data, allowing pilots and engineer to inhabit the remote subsea environment. Verbal directives function as a key mode of control, where the inspector’s directives (action) prompt specific ROV movements by the pilot, which in turn change the visual input the inspector receives (perception), influencing their next directive. The pilot’s seamless, embodied responses, where verbal agreement is inseparable from ROV manipulation, highlight

the integration of cognition and action in cyborgenic work. As Clark proposes, our cognitive processes are inherently intertwined with the tools and technologies we employ. These processes do not lead human workers to become essentially different but rather expands the reach and capabilities of human work and creativity. This study reveals workplace cyborgification as a dynamic process, not a static state, where human-technology integration evolves through distributed action-perception loops. Onshoring exemplifies how cyborgification in the ROC reshapes not only *where* work gets done but also some of the microdynamics of *how* human capabilities get extended.

Our study offers three contributions to CSCW. First, the notion of workplace cyborgification may seem provocative, but it provides a tangible way to analyze the increasingly tight entanglements between humans and technology in complex work settings, moving us beyond metaphorical uses of the cyborg-term. Secondly, our ethnographic data reveals the nuanced pragmatic strategies (including speech acts) and shared situational awareness arising as workers engage in cyborg labor for high-stakes, technologically mediated, remote work. Third, we empirically detail how modern control rooms are shifting from centralized monitoring stations to distributed systems of action, integrating human operators into cybernetic loops. Maritime onshoring represents a significant transformation of work, not just in terms of location, but also in the fundamental ways that stakeholders perceive, interact with, and exert control over their work environment. In broader terms, onshoring reshapes tasks, roles, professional identities, and workforce composition — enabling, for instance, participation by people unable to go offshore. This has deep implications for the maritime workplace. Yet, reconciling the onshore model with the preferences of personnel who value the social dynamics of working at sea is likely to be a challenge.

6 Conclusion

Viewing maritime onshoring through the lens of workplace cyborgification reveals a tapestry of extended perception, remote agency, and distributed cognition. Through a technologically-mediated sensorium and tightly co-operative actions, pilots and inspectors become human-technology hybrids, seamlessly translating speech and sensation into precise remote action in the subsea environment. Through its conceptual framing this ethnographic case study highlights a need for CSCW researchers to engage more deeply with the rich empirical and conceptual dimensions of workplace cyborgification, in maritime oil and gas, as well as other domains

of remote work. This includes studying the situated pragmatics of speech acts and other modalities through which human and non-human participants coordinate actions, manage human-machine interfaces, and fulfill institutional demands.

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